



NORTH SYDNEY BOYS HIGH SCHOOL

2016 HSC ASSESSMENT TASK 2

Mathematics

General Instructions

Reading Time – 5 minutes
Working Time – 60 minutes
Total Time – 65 minutes

Questions are to be answered in the booklet provided, showing ALL necessary working.

- Write using blue or black pen
- Board approved calculators may be used
- Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher: Please tick

- ☐ Mr Berry
- ☐ Mr Hwang
- ☐ Mr Ireland
- ☐ Dr Jomaa
- ☐ Ms Lee
- ☐ Ms Ziaziaris
- ☐ Mr Zuber

Student Number: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	5	Total	Total %
Mark	$\frac{\quad}{10}$	$\frac{\quad}{8}$	$\frac{\quad}{7}$	$\frac{\quad}{9}$	$\frac{\quad}{8}$	$\frac{\quad}{42}$	$\frac{\quad}{100}$

Question 1 (10 Marks)**Marks**

(a) Find the primitive of each of the following:

(i) $\int (2x^3 - 3x^2 + 1) dx$ **2**

(ii) $\int (2 - 3x)^5 dx$ **2**

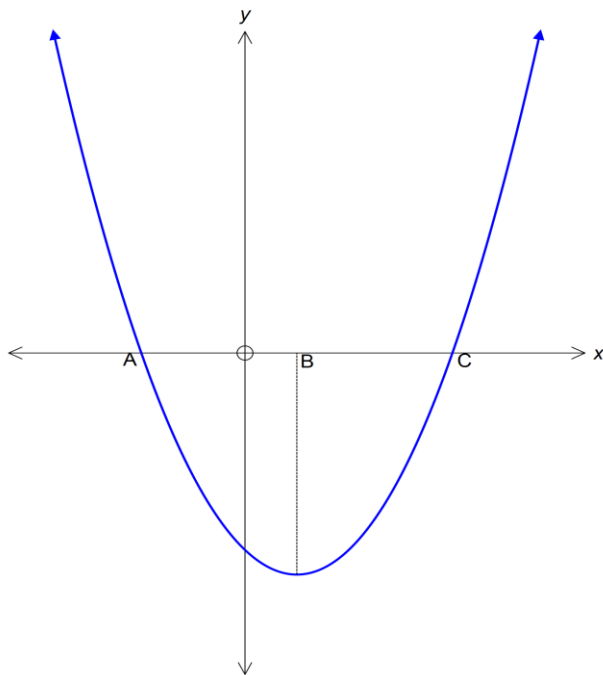
(b) Evaluate:

(i) $\int_1^2 \frac{t^4 + 1}{t^2} dt$ **2**

(ii) $\int_1^4 \frac{dx}{\sqrt{x}}$ **2**

(c)

Sketch a possible primitive of the function shown below:

2

Question 2 (8 Marks)

- (a) Evaluate

$$\int_0^2 (e^{3x} + 1) dx$$

2

- (b) Differentiate

$$y = \frac{x^2 + e^x}{x - 1}$$

3

- (c) Find the equation of the tangent to the curve $y = e^{3x}$ at the point whose x coordinate is 2.

3

Question 3 (7 Marks)

- (a) Consider the function $y = x^2 e^x$:

(i) For what values of x is this function defined?

1

(ii) Describe the behavior of the function as x :

α) approaches $+\infty$

1

β) approaches $-\infty$

1

(iii) Find any stationary points and determine their nature.

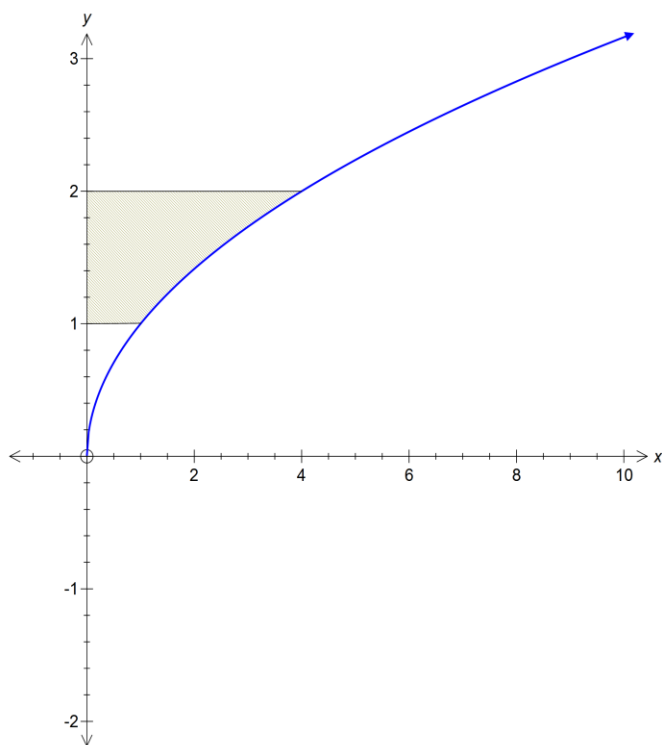
3

(iv) Sketch the curve of this function on the number plane.

1

Question 4 (9 Marks)

(a)



The region above is the area between $y = \sqrt{x}$, $y = 1$, $y = 2$ and the y -axis.

Find the exact area of the shaded region.

3

(b)

(i) Copy and complete the following table:

1

x	-2	-1	0	1	2
$4 - x^2$					

(ii) Hence estimate $\int_{-2}^2 (4 - x^2) dx$ using four applications of the trapezoidal rule.

2

(iii) Find the exact value of $\int_{-2}^2 (4 - x^2) dx$

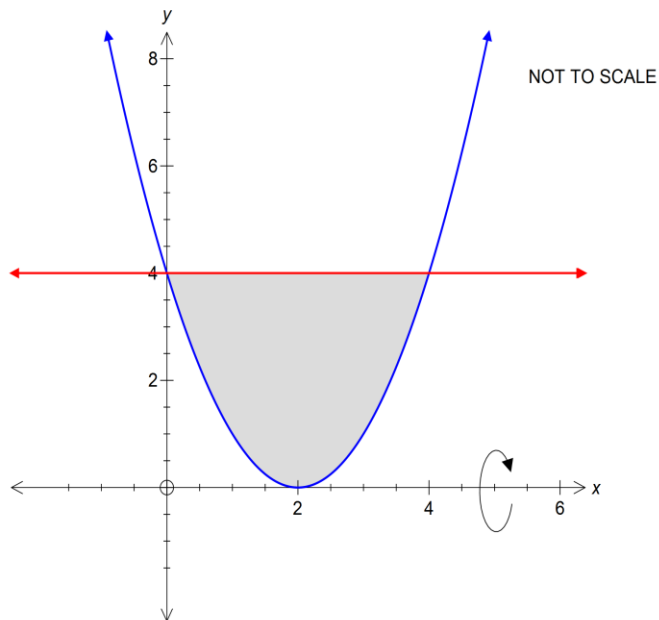
2

(iv) Did you over estimate or under estimate the exact integral in (ii)? Give a reason for your answer.

1

Question 5 (8 Marks)

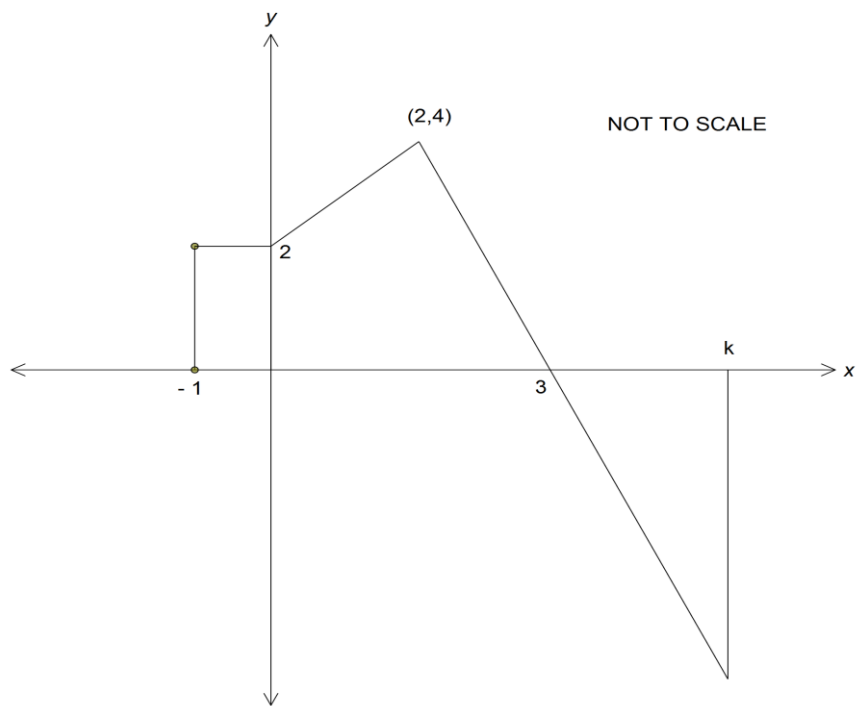
- (a) The shaded region bounded by the graph $y = (x - 2)^2$ and the line $y = 4$ is rotated about the x-axis to form a solid of revolution as show in the diagram.



Find the volume of the solid formed. Leave your answer in exact form.

4

- (b) Consider the graph $y = f(x)$ below.



If $\int_{-1}^k f(x) dx = 0$ find the value of k .

4

Question 1

(a) (i)

$$\int (2x^3 - 3x^2 + 1) dx = \frac{x^4}{2} - \frac{x^3}{3} + x + C$$

✓ -1 for each error
✓

(a) (ii)

$$\begin{aligned} \int (2 - 3x)^5 dx &= \frac{(2 - 3x)^6}{-3 \times 6} + C \\ &= -\frac{(2 - 3x)^6}{18} + C \end{aligned}$$

✓ Correct integral

✓ Simplified solution

(b) (i)

$$\begin{aligned} \int_1^2 \frac{t^4 + 1}{t^2} dt &= \int_1^2 \left(t^2 + \frac{1}{t^2} \right) dt \\ &= \left[\frac{t^3}{3} - \frac{1}{t} \right]_1^2 \\ &= \left(\frac{8}{3} - \frac{1}{2} \right) - \left(\frac{1}{3} - 1 \right) \\ &= \frac{17}{6} \end{aligned}$$

✓ Correct integral

✓ Solution

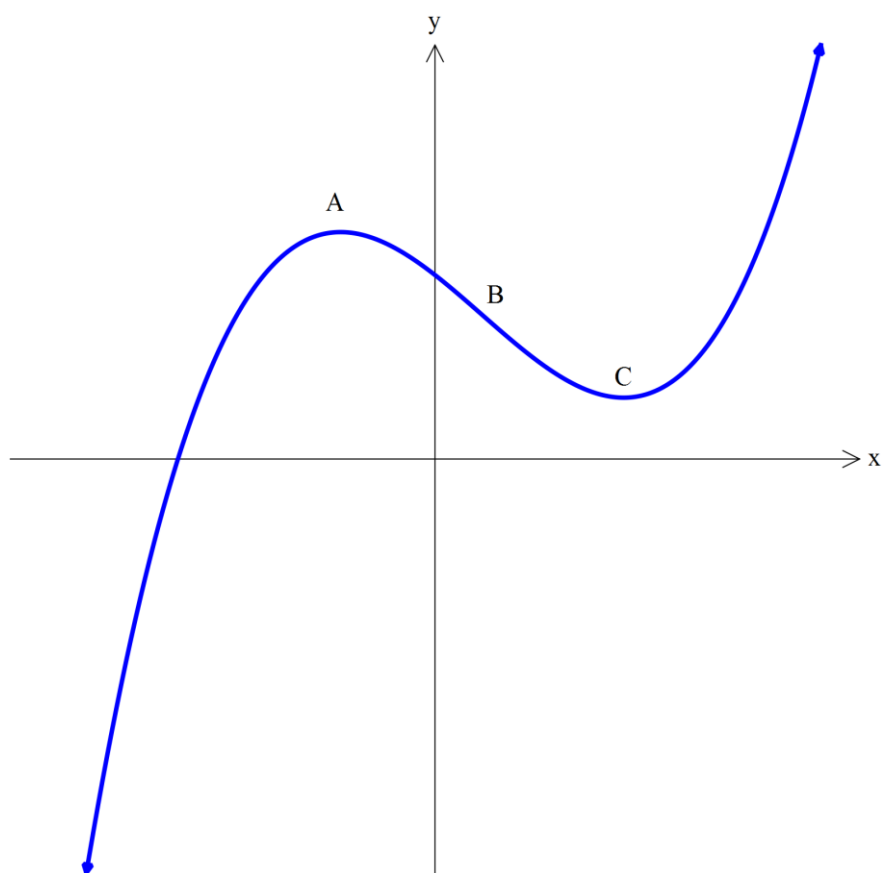
(b) (ii)

$$\begin{aligned} \int_1^4 \frac{dx}{\sqrt{x}} &= \int_1^4 x^{-\frac{1}{2}} dx \\ &= \left[\frac{1}{2} x^{\frac{1}{2}} \right]_1^4 \\ &= \frac{1}{2} (\sqrt{4} - \sqrt{1}) \\ &= \frac{1}{2} \end{aligned}$$

✓ Correct integral

✓ Solution

(c)



- ✓ Correct shape
- ✓ Points in correct positions

note the vertical translation does not matter

Question 2

(a)

$$\int_0^2 (e^{3x} + 1) dx = \left[\frac{e^{3x}}{3} + x \right]_0^2$$

$$= \left(\frac{e^6}{3} + 2 \right) - \left(\frac{1}{3} + 0 \right)$$

$$= \frac{e^6 + 5}{3}$$

✓ Integral

✓ Solution

(b)

$$\text{let } y = \frac{u}{v}$$

$$\therefore u = x^2 + e^x$$

$$u' = 2x + e^x$$

$$v = x - 1$$

$$v' = 1$$

$$y' = \frac{u'v - v'u}{v^2}$$

$$= \frac{(2x + e^x)(x - 1) - (1)(x^2 + e^x)}{(x - 1)^2}$$

$$= \frac{2x^2 - 2x + xe^x - e^x - x^2 - e^x}{(x - 1)^2}$$

$$= \frac{x^2 + xe^x - 2e^x - 2x}{(x - 1)^2}$$

✓ Quotient rule

✓ Substitution into formula

✓ Final answer

(c)

$$y = e^{3x}$$

$$\frac{dy}{dx} = 3e^{3x}$$

when $x = 2$

$$m = 3e^6$$

and $y = e^6$

$$y - y_1 = m(x - x_1)$$

$$y - e^6 = 3e^6(x - 2)$$

$$y - e^6 = 3e^6x - 6e^6$$

$$\therefore y = 3e^6x - 5e^6$$

or

$$3e^6x - y - 5e^6 = 0$$

✓ Gradient

✓ Substitution into formula

✓ Final answer

Question 3

(a) (i) All real x

✓ Answer

$$(ii) (\alpha) \lim_{x \rightarrow \infty} x^2 e^x = \infty$$

✓ Answer

$$(\beta) \lim_{x \rightarrow -\infty} x^2 e^x = 0$$

✓ Answer

(a) (iii)

$$\text{let } y = uv$$

$$u = x^2$$

$$u' = 2x$$

$$v = e^x$$

$$v' = e^x$$

$$y' = u'v + v'u$$

$$= (2x)(e^x) + (x^2)(e^x)$$

$$= xe^x(x + 2)$$

✓ Derivative

$$\text{when } y' = 0$$

$$xe^x(x + 2) = 0$$

$$\therefore x = -2 \text{ or } x = 0$$

$\therefore$ stationary points occur at $\left(-2, \frac{4}{e^2}\right)$ and $(0,0)$

✓ Stationary points

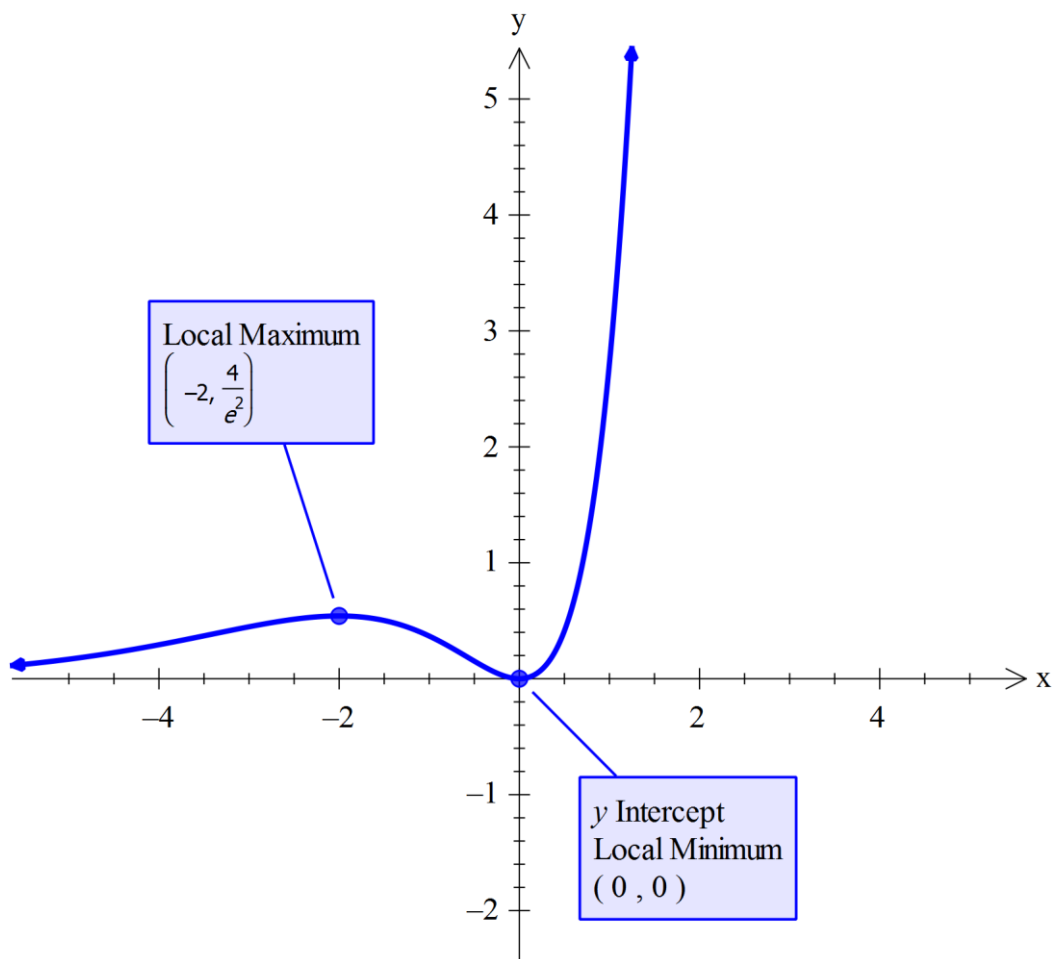
test the nature:

x	-3	-2	-1	0	1
y'	$+\frac{3}{e^3}$	0	$-\frac{1}{e}$	0	$+3e$

$\therefore (-2, \frac{4}{e^2})$ is a local maxima and $(0,0)$ is a local minima.

✓ Testing nature

(a) (iv)



✓ Correct graph with details labelled

Question 4

(a) (i)

$$y = \sqrt{x}$$

$$\therefore x = y^2$$

$$A = \int_1^2 x \, dy$$

$$= \int_1^2 y^2 \, dy$$

$$= \left[\frac{y^3}{3} \right]_1^2$$

$$= \frac{8}{3} - \frac{1}{3}$$

$$= \frac{7}{3} \text{ units}^2$$

✓ Making x the subject

✓ Correct integral

✓ Correct answer (must say units²).

(b) (i)

x	-2	-1	0	1	2
$4 - x^2$	0	3	4	3	0

✓ Correct table

(b) (ii)

$$A \approx \frac{h}{2} (y_0 + y_4 + 2(y_1 + y_2 + y_3))$$

$$= \frac{1}{2} (0 + 0 + 2(3 + 4 + 3))$$

$$= 10 \text{ units}^2$$

✓ Trapezoidal rule

✓ Answer

(b) (iii)

$$A = \int_{-2}^2 (4 - x^2) \, dx$$

$$= \left[4x - \frac{x^3}{3} \right]_{-2}^2$$

$$= \left((4)(2) - \frac{8}{3} \right) - \left((4)(-2) - \frac{(-8)}{3} \right)$$

$$= \frac{24}{3} - \frac{8}{3} + \frac{24}{3} - \frac{8}{3}$$

$$= \frac{32}{3} \text{ units}^2$$

✓ Correct integral

✓ Correct answer

(b) (iv)

Underestimated because the function is concave down.

✓ Must give valid reason

Question 5

(a)

First find points of intersection

$$\begin{aligned} y &= (x-2)^2 \dots \textcircled{1} \\ y &= 4 \dots \textcircled{2} \\ \textcircled{1} &= \textcircled{2} \end{aligned}$$

$$\begin{aligned} \therefore (x-2)^2 &= 4 \\ x^2 - 4x + 4 &= 4 \\ x^2 - 4x &= 0 \\ x(x-4) &= 0 \\ \therefore x &= 0 \text{ or } x = 4 \end{aligned}$$

✓ Intercepts

Subtract the volumes under the parabola from the volume of the cylinder

$$\begin{aligned} V &= \pi r^2 h - \left(\pi \int_2^4 ((x-2)^2)^2 dx + \pi \int_0^2 ((x-2)^2)^2 dx \right) \\ &= \pi(4^2)(4) - \pi \int_0^4 (x-2)^4 dx \\ &= 64\pi - \pi \left[\frac{(x-2)^5}{5} \right]_0^4 \\ &= 64\pi - \pi \left(\frac{2^5}{5} - \frac{(-2)^5}{5} \right) \\ &= 64\pi - \frac{64\pi}{5} \\ &= \frac{241\pi}{5} \text{ units}^3 \end{aligned}$$

✓ Forming equation

✓ Substituting values into integrals/formula

✓ Answer

(b)

$$\begin{aligned}\int_{-1}^3 f(x) dx &= \text{Rectangle} + \text{Trapezium} + \text{Triangle} \\ &= (1 \times 2) + \frac{2}{2}(2 + 4) + \left(\frac{1}{2}\right)(1)(4) \\ &= 10 \\ \therefore \int_3^k f(x) dx &= 10\end{aligned}$$

✓ Area above x axis

To find the area under the x-axis in terms of k .

$$\begin{aligned}y &= -4x + 12 \\ \text{when } x &= k \\ y &= -4k + 12\end{aligned}$$

$$\int_3^k f(x) dx = -10$$

✓ Area below axis the same magnitude

$$\therefore \left(\frac{1}{2}\right)(k-3)(12-4k) = -10$$

✓ Setting up equation

$$\begin{aligned}(k-3)(6-2k) &= -10 \\ 6k - 2k^2 - 18 + 6k &= -10 \\ 2k^2 - 12k + 8 &= 0 \\ k^2 - 6k + 4 &= 0\end{aligned}$$

$$k = \frac{6 \pm \sqrt{(-6)^2 - (4)(4)(1)}}{(2)(1)}$$

$$\therefore k = \frac{6 + \sqrt{20}}{2} \quad (\text{since } k > 3)$$

$$= \frac{6 + 2\sqrt{5}}{2}$$

$$= 3 + \sqrt{5}$$

✓ Answer